EXPERIMENT-3

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1. COMMAND LINE INTERFACE: PYTHON CODE:

```
import os

import sys

def rename_file(old_name, new_name):

try:

os.rename(old_name, new_name)

print(f"File renamed from {old_name} to {new_name}")

except FileNotFoundError:

print(f"Error: {old_name} not found.") except Exception as e:

print(f"An error occurred: {e}")

if __name___== "__main__": if len(sys.argv) != 3:

print("Usage: python rename_file_cli.py ") else: rename_file(sys.argv[1],
sys.argv[2])
```

OUTPUT:

```
D:\230701304>python rename_file_cli.py demo.txt shal.txt
File renamed from demo.txt to shal.txt
```

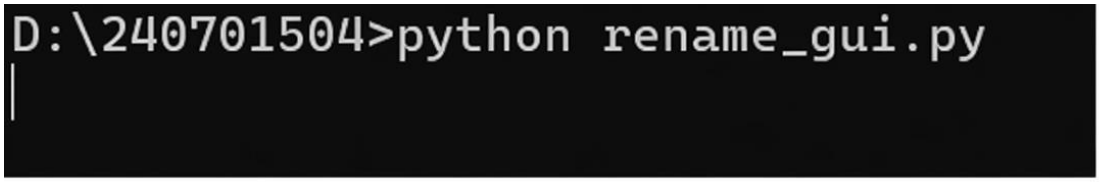
2. GRAPHICAL USER INTERFACE:

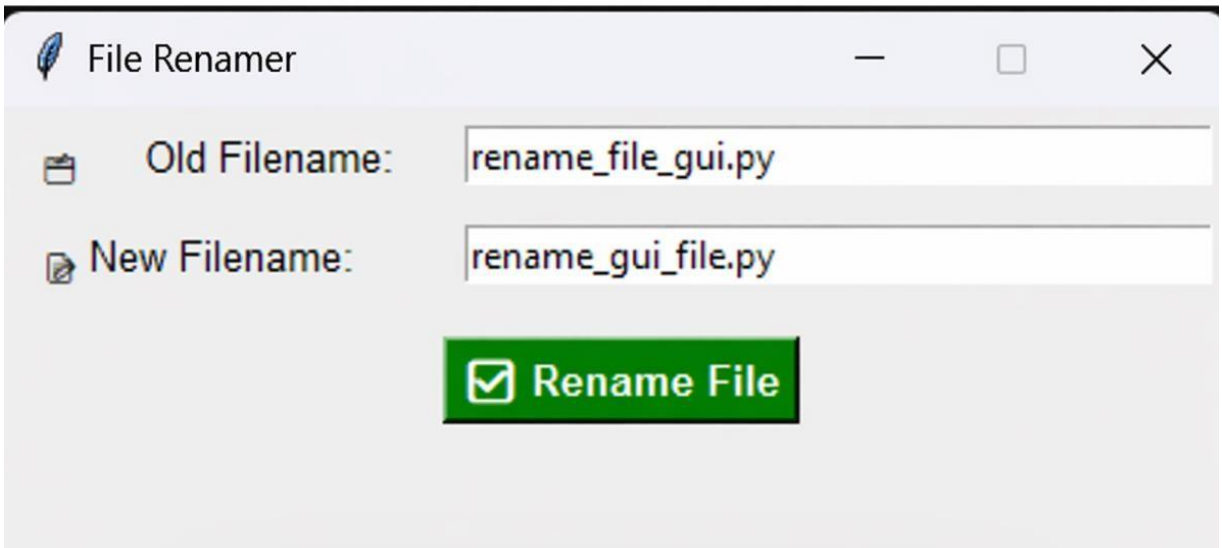
PYTHON CODE:

```
import tkinter as tk

from tkinter import messagebox

import os

# Function to rename the file
```

```
def rename_file():

old_name = old_filename_entry.get().strip() new_name =
new_filename_entry.get().strip()

# Check if input fields are empty if not old_name or not

new_name: messagebox.showwarning("Warning", "Please enter both
filenames!")

return

# Check if the old file exists if not os.path.exists(old_name):
messagebox.showerror("Error", f"File '{old_name}' not found.")

return

# Check if the new file already exists if os.path.exists(new_name): overwrite =
messagebox.askyesno("Warning", f"'{new_name}' already exists. Overwrite?")

if not overwrite: return try: os.rename(old_name, new_name)
messagebox.showinfo("Success", f"File renamed from '{old_name}' to
'{new_name}'")

except Exception as e: messagebox.showerror("Error", f"An error occurred:
{e}")

 # Create main window root = tk.Tk() root.title("File Renamer")
root.geometry("400x150") root.resizable(False, False)

 # Fixed window size # Labels tk.Label(root, text="Old Filename:",
font=("Arial", 10)).grid(row=0, column=0, padx=10, pady=5, sticky="w")
tk.Label(root, text="New Filename:", font=("Arial", 10)).grid(row=1,
column=0, padx=10, pady=5, sticky="w")

# Entry fields old_filename_

entry = tk.Entry(root, width=40) old_filename_entry.grid(row=0, column=1,
padx=10, pady=5)

new_filename_entry = tk.Entry(root, width=40)
new_filename_entry.grid(row=1, column=1, padx=10, pady=5)

 # Styled Button
```

rename_button = tk.Button(root, t e x t = " ✅ Rename File", bg="green", fg="white", font=("Arial", 10, "bold"), command=rename_file) rename_button.grid(row=2, column=0, columnspan=2, pady=10) # Run the GUI event loop root.mainloop()





3.VOICE USER INTERFACE:

PYTHON CODE:

import speech_recognition as sr

import os

def rename_file_from_voice_command(command):

try: words = command.lower().split(" ")

if "rename" in words and "to" in words:

rename_index = words.index("rename") to_index = words.index("to")

Extract old and new filenames old_name = words[rename_index + 1] new_name = words[to_index + 1]

Check if file exists

if not os.path.exists(old_name):

print(f"Error: File '{old_name}' not found.")

return

Rename file os.rename(old_name, new_name)

print(f"✅ File renamed from '{old_name}' to '{new_name}'")

else:

print("Invalid command format. Say: 'Rename oldfile.txt to newfile.txt'")

except Exception as e:

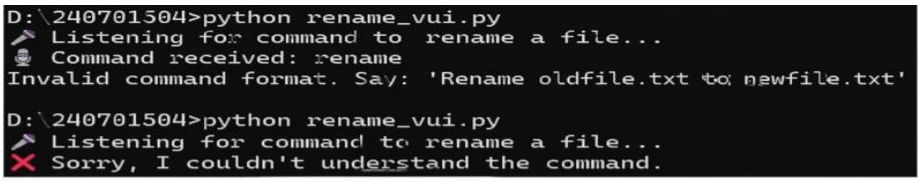
print(f"Error: {e}") def listen_for_command(): recognizer = sr.Recognizer() mic = sr.Microphone()

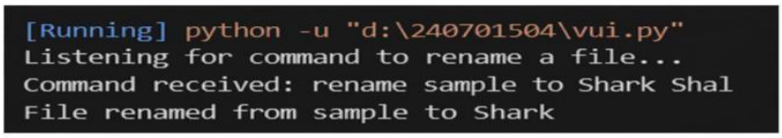
print("🎤 Listening for command to rename a file...") with mic as source: recognizer.adjust_for_ambient_noise(source) audio = recognizer.listen(source) try:

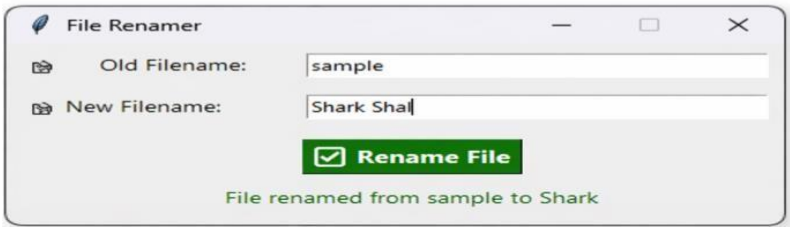
command = recognizer.recognize_google(audio)

print(f"🗣 Command received: {command}") rename_file_from_voice_command(command) except sr.UnknownValueError: print("❌ Sorry, I couldn't understand the command.") except sr.RequestError as e:

print(f"⚠ Could not request results from Google Speech Recognition service; {e}") if __name__== "__main__": listen_for_command()







4. USER SATISFACTION COMPARISON:

PYTHON CODE:

```
survey():
 print("Rate your satisfaction with the following interfaces (1-5):")
 # Get user input for each interface
try:
 cli_satisfaction = int(input("CLI (1-5): "))
gui_satisfaction = int(input("GUI (1-5): "))
vui_satisfaction = int(input("VUI (1-5): "))
# Ensure valid ratings
if not (1 <= cli_satisfaction <= 5 and 1 <= gui_satisfaction <= 5 and 1 <=
vui_satisfaction <= 5):
 print("Please enter ratings between 1 and 5 only.")
return
#Display the ratings print("\nYour satisfaction ratings:")
 print(f"CLI: {cli_satisfaction}")
 print(f"GUI: {gui_satisfaction}")
 print(f"VUI: {vui_satisfaction}")
# Calculate the average satisfaction
avg_satisfaction = (cli_satisfaction + gui_satisfaction + vui_satisfaction) / 3
print(f"\nAverage Satisfaction Score: {avg_satisfaction:.2f}")
except ValueError: print("Invalid input! Please enter numbers between 1 and
5.")
 # Run the survey function if __name___== "__main__": survey()
```

```
D:\240701504>python survey.py
Rate your satisfaction with the following interfaces (1-5):
CLI (1-5): 4
GUI (1-5): 5
VUI (1-5): 3

Your satisfaction ratings:
CLI: 4
GUI: 5
VUI: 3

Average Satisfaction Score: 4.00

D:\240701504>
```